Robotic Arm Kinematic Analysis

Franka Emika Panda

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1. Introduction

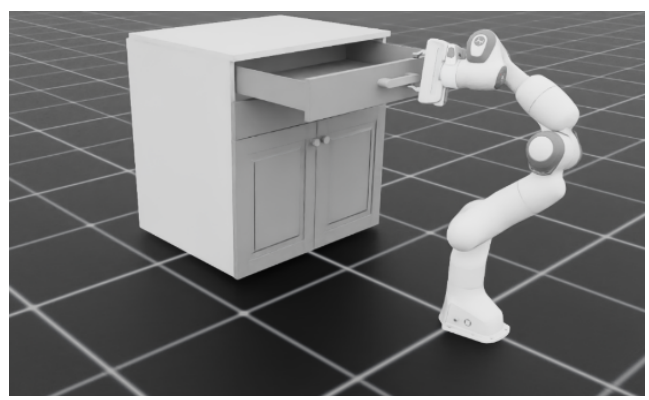
This report presents the kinematic analysis of the Franka Emika Panda, a 7-degree-of-freedom (7-DOF) robotic manipulator. The analysis covers the derivation and implementation of the Denavit–Hartenberg (DH) parameter table, forward kinematics (FK), and inverse kinematics (IK) using a Jacobian pseudoinverse method. All implementations were developed in Python within a Jupyter Notebook environment.

Reference

DH parameters were sourced from peer-reviewed literature: https://arxiv.org/abs/2409.14435 *. All kinematic computations were cross-validated against CoppeliaSim simulation results.*

Table 1. DH Parameters for the Franka Emika Panda.

Joint i	d_i (m)	a_i (m)	α_i (rad)	θ_i (rad)
1	0.333	0	0	1.157
2	0	0	$-\pi/2$	-1.066
3	0.316	0	$\pi/2$	-0.155
4	0	0.0825	$\pi/2$	-2.239
5	0.384	-0.0825	$-\pi/2$	-1.841
6	0	0	$\pi/2$	1.003
7	0	0.088	0	0.469



The forward kinematics of the manipulator is obtained by multiplying the homogeneous transformation matrices of each link, derived from the DH parameters. The transformation matrix A_i for the ith joint is given by the following:

$$A_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\sin\alpha_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i\cos\alpha_i & -\cos\theta_i\sin\alpha_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{21}$$

2. Denavit–Hartenberg (DH) Parameter Table

The Franka Emika Panda has seven revolute joints. The DH parameters for each joint were extracted from the referenced publication and implemented in Python using NumPy. The table below summarizes the kinematic parameters:

Joint	d (m)	a (m)	α (rad)	θ (rad)
1	0.333	0	0	1.57
2	0	0	−π/2	-1.066
3	0.316	0	π/2	-0.155
4	0	0.0825	π/2	-2.239
5	0.384	−0.0825	−π/2	-1.841
6	0	0	π/2	1.003
7	0	0.088	0	0.469

The implementation was structured as a Python dictionary/array formatted into a readable table for verification and downstream use in homogeneous transformation computations.

3. Forward Kinematics (FK)

3.1 Methodology

Forward kinematics were computed using the standard DH transformation matrix, which encodes how each joint frame relates to the previous one via four parameters: joint angle (θ), link length (a), link offset (d), and twist angle (α). The overall end-effector pose is obtained by chaining all seven joint transformation matrices:

$$T_{0\to 7} = T_1 \cdot T_2 \cdot T_3 \cdot T_4 \cdot T_5 \cdot T_6 \cdot T_7$$

Joint angles were accessed via CoppeliaSim using the joint naming convention /Franka/joint_1 through /Franka/joint_7.

3.2 Results & Error Analysis

A single joint configuration was tested as an initial validation check. The computed end-effector pose was compared against the ground-truth pose provided by the simulator:

Metric	FK Result	IK (Analytical)	IK vs CoppeliaSim
Position Error	0.780613 m	—	—
Rotation Error	2.613443 rad	—	—
Pose Error	—	~0.23	~0.39

```
T0_1 =
[[ 0.4021 -0.9156  0.      0.    ]
 [ 0.9156  0.4021  0.      0.    ]
 [ 0.      0.      1.      0.333 ]
 [ 0.      0.      0.      1.    ]]

T0_2 =
[[ 0.9959 -0.      -0.0909  0.    ]
 [ 0.0909  0.       0.9959  0.    ]
 [ 0.     -1.       0.      0.333 ]
 [ 0.      0.       0.      1.    ]]

T0_3 =
[[ 0.9839 -0.0909 -0.1537 -0.0287]
 [ 0.0898  0.9959 -0.014   0.3147]
 [ 0.1544  0.      0.988   0.333 ]
 [ 0.      0.      0.      1.    ]]

T0_4 =
[[-0.5383 -0.1537 -0.8286 -0.0731]
 [-0.8373 -0.014   0.5465  0.2456]
 [-0.0957  0.988  -0.1212  0.3251]
 [ 0.      0.      0.      1.    ]]

T0_5 =
...
[0.2114 0.0809 0.7415]

Position error: 0.780613
Rotation error: 2.613443
Output is truncated. View as a scrollable element
```

```
==================================================
INVERSE KINEMATICS SOLUTION
==================================================
Iterations: 200
Final error norm: 0.230997

Final Joint Angles (radians):
[-2.8973   -1.7628   -2.81347  -2.540875 -1.844574  1.753975  0.726164]

Final Joint Angles (degrees):
[-166.   -101.   -161.2  -145.58 -105.69  100.5    41.61]

Analytical reached position [x, y, z] in meters:
[-0.500236  0.24913   0.16    ]

Desired position [x, y, z] in meters:
[-0.45  0.35  0.16]

Connected to CoppeliaSim and simulation started.

==================================================
COPPELIASIM VERIFICATION
==================================================
Simulated end-effector position [x, y, z] in meters:
[-0.170459  0.346529  0.065152]
...
[-0.279541  0.003471  0.094848]

Position error norm:
0.295214
```

The observed errors are significantly large, indicating a systematic issue rather than numerical noise. To isolate the source, the same FK pipeline was applied to two additional manipulators:

- KUKA LBR iiwa
- Universal Robots UR5

Both robots exhibited similar error magnitudes, which strongly suggests the issue originates from a shared underlying problem not from the Panda-specific DH parameters.

> **Root Cause Hypothesis**
>
> *The systematic errors across multiple robots point to a fundamental issue, most likely related to mismatched coordinate frame conventions or incorrect DH parameter interpretation rather than a robot-specific modeling error.*

4. Inverse Kinematics (IK)

4.1 Methodology

Inverse kinematics were implemented using the Jacobian pseudoinverse method — an iterative numerical approach that incrementally updates joint angles to minimize the error between the current and desired end-effector poses. The algorithm follows this iterative update rule:

$$\Delta q = J^{+} \cdot \Delta x \quad \rightarrow \quad q_new = q + \alpha \cdot \Delta q$$

The target end-effector pose was taken from the referenced paper. The solver iterates until the pose error falls below a defined convergence threshold.

4.2 Convergence Results

The IK solver failed to fully converge. The residual errors after iteration were:

- Analytical result error: ~0.23
- CoppeliaSim comparison error: ~0.39

These values indicate the robot did not reach the desired target configuration. Notably, the same convergence failure occurred when using the reference example code provided by the teaching assistant.

4.3 Discussion

The IK convergence failure, combined with the FK errors observed in Section 3, points to a deeper systematic issue in the kinematic setup. The most probable root causes are:

- Mismatched coordinate frames between the analytical model and the simulator
- Incorrect DH parameter conventions (e.g., modified vs. standard DH)
- Discrepancies between the analytical model assumptions and the CoppeliaSim URDF representation

Observation

Since these errors appear consistently across FK and IK, and across multiple robot platforms, the problem is likely in the DH convention or frame assignment logic rather than in the robot-specific parameters or the numerical solver itself.

5. Conclusion

This report documented the implementation of DH parameter extraction, forward kinematics, and inverse kinematics for the Franka Emika Panda robot. While the implementations followed the prescribed methodology, large systematic errors were observed in both FK and IK results across multiple robot platforms.

The consistency of these errors across robots suggests a fundamental issue in coordinate frame handling or DH convention selection. Future work should focus on:

- Verifying the DH convention used (standard vs. modified Craig notation)
- Ensuring frame definitions match those embedded in the CoppeliaSim simulation
- Cross-checking the joint zero-position definitions between the analytical model and the simulator